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Original Article

Postoperative Ketoacidosis in Cardiac Surgery: Incidence, Analytical Profile, and Association With Sodium-Glucose Cotransporter 2 Inhibitors—A Prospective Observational Study

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Objective: To determine the incidence of postoperative ketoacidosis after elective cardiac surgery and to evaluate its association with sodium-glucose cotransporter 2 (SGLT2) inhibitor therapy and perioperative biochemical parameters.
Design: A prospective observational study.
Setting: Cardiac intensive care unit of a tertiary academic hospital in Madrid, Spain, between July 2024 and May 2025.
Participants: Adult patients undergoing elective cardiac surgery. Patients undergoing emergency procedures or those without informed consent were excluded.
Interventions: Standard perioperative care. Arterial blood gases and capillary ketone measurements were obtained at intensive care unit (ICU) admission (defined as postoperative hour 0), at postoperative hour 6, and at postoperative hour 16.
Measurements and Main Results: Postoperative ketoacidosis was defined as arterial pH <7.30 and ketonemia ≥0.6 mmol/L in the absence of preoperative ketoacidosis. A total of 107 patients were included (28 SGLT2 inhibitor users and 79 nonusers). Postoperative ketoacidosis occurred in 17 of 28 patients (54.8%) receiving SGLT2 inhibitors, compared with 8 of 79 (10.1%) in the non-SGLT2 inhibitor group ($p < 0.001$). Diabetes mellitus was independently associated with ketoacidosis (odds ratio, 5.13; $p = 0.001$). Patients with ketoacidosis had lower arterial pH (7.31 *v* 7.35, $p < 0.001$), lower bicarbonate (21.6 *v* 22.8 mmol/L, $p = 0.012$), and more negative base excess (−4.2 *v* −2.6, $p = 0.03$). No significant differences were observed in ICU length of stay, duration of mechanical ventilation, reintervention, or mortality.
Conclusions: Postoperative ketoacidosis, frequently euglycemic, is relatively common after cardiac surgery and strongly associated with SGLT2 inhibitor therapy. Careful monitoring and tailored perioperative management should be considered for this high-risk group.
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Key Words: cardiac surgery; postoperative complications; euglycemic diabetic ketoacidosis; sodium-glucose cotransporter 2 inhibitors; critical care; intensive care unit

SODIUM-GLUCOSE COTRANSPORTER 2 (SGLT2) inhibitors have become a cornerstone in the management of type 2 diabetes mellitus and heart failure, owing to their ability to improve glycemic control while conferring significant cardiorenal protection. Their expanding indications have led to widespread use among patients undergoing cardiac surgery. However, a growing number of reports have described euglycemic diabetic ketoacidosis (euDKA) as a rare but potentially serious adverse effect, particularly in the context of fasting, surgical stress, or critical illness.[1,2] EuDKA is characterized by metabolic acidosis with elevated ketones and normal or

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only mildly elevated glucose levels, which may delay diagnosis and treatment, allowing metabolic disturbances to progress unnoticed.[3]

Cardiac surgery imposes profound metabolic and endocrine stress through mechanisms including prolonged fasting, catecholamine surge, exposure to extracorporeal circulation, and the frequent occurrence of postcardiotomy low cardiac output states. These factors, together with insulin resistance, inflammation, and altered substrate utilization, create a milieu prone to ketone body formation and acid-base imbalance.[4-6] Patients with diabetes or heart failure treated with SGLT2 inhibitors may therefore be at particularly high risk, even when standard glycemic control is achieved. While current guidelines recommend withholding SGLT2 inhibitors before noncardiac surgery to minimize the risk of euDKA,[7,8] evidence in the cardiac surgical population remains limited, and most published data derive from retrospective series or isolated case reports.[9,10]

The present prospective observational study was designed to determine the incidence of postoperative ketoacidosis in patients undergoing elective cardiac surgery and to explore its association with perioperative SGLT2 inhibitor therapy. Secondary objectives were to identify clinical, pharmacological, and analytical variables related to its development and to assess whether the presence of postoperative ketoacidosis is associated with worse clinical outcomes—including intensive care unit (ICU) length of stay, duration of mechanical ventilation, surgical reintervention, or in-hospital mortality—as well as with alterations in key biochemical parameters such as lactate, base excess, and acid-base balance.

Methods

Study Design and Setting

We conducted a prospective observational study in a cardiac ICU of a tertiary academic center in Madrid, Spain. The study included all consecutive adult patients admitted to the ICU following elective cardiac surgery between July 2024 and May 2025. The study was conducted in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology guidelines.[11]

Ethical Approval

Ethical approval for this study (protocol number 24/344) was provided by the Clinical Research Ethics Committee of Hospital Universitario 12 de Octubre, Madrid, Spain (Chairperson Dr. Dulce María Ugalde Diez), on June 25, 2024.

Inclusion and Exclusion Criteria

Inclusion criteria were age ≥18 years and elective cardiac surgery (valvular, coronary, aortic, or combined procedures). Patients who underwent emergency surgery or did not sign informed consent were excluded.

Variables and Data Collection

Demographic data (age, sex, weight, height, body mass index [BMI]), comorbidities (hypertension, diabetes mellitus, dyslipidemia), chronic medication (including SGLT2 inhibitors), surgical variables (type of surgery, cardiopulmonary bypass use and time, EUROSCORE II), and outcomes (ICU stay, mechanical ventilation duration, need for surgical reintervention, in-hospital mortality) were collected prospectively.

To ensure accurate diagnosis of postoperative ketoacidosis, a structured sampling protocol was prospectively applied. Capillary ketonemia and arterial blood gases were obtained at five key time points. First, upon admission to the cardiothoracic ward the day before surgery (baseline), a ketone measurement was performed to rule out preexisting ketosis. Second, a preoperative sample was collected at 7 AM on the day of surgery, after overnight fasting, to exclude fasting-induced ketosis. Postoperatively, arterial blood gases and capillary ketonemia were obtained at 3 fixed time points: at ICU admission (postoperative hour 0), at postoperative hour 6, and at postoperative hour 16. These time points coincide with the standardized unit protocol for blood gas and laboratory analyses, ensuring uniform and reproducible data collection, and are spaced sufficiently to detect ketoacidosis during the immediate postoperative period (Fig 1).

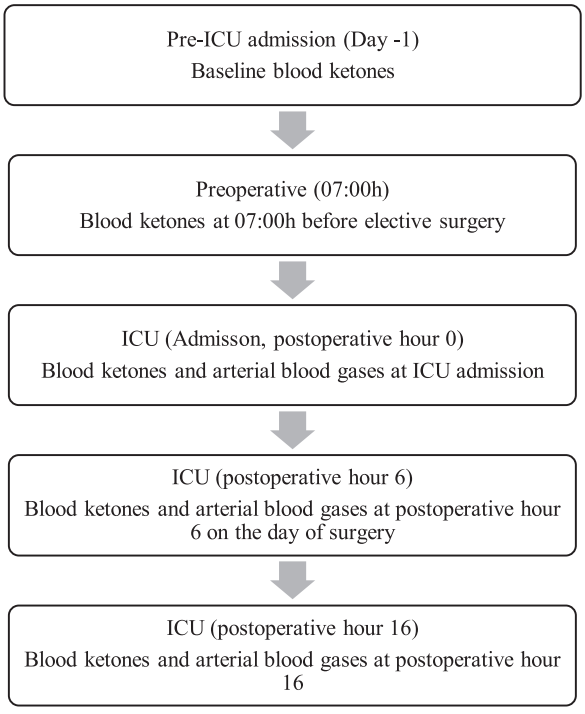


Fig 1. Sample collection protocol timeline. Schematic representation of the structured sampling protocol used for perioperative monitoring of arterial blood gases and capillary ketonemia. Baseline measurement was performed the day before surgery; preoperative samples were obtained after overnight fasting (7 AM). Postoperative samples were taken at intensive care unit admission (postoperative hour 0), at postoperative hour 6, and at postoperative hour 16, coinciding with routine laboratory analyses.

Perioperative Management of SGLT2 Inhibitors

During the study period, perioperative discontinuation protocols for SGLT2 inhibitors were not yet standardized. In our institution, elective cardiac surgery patients were admitted to the cardiothoracic ward on the afternoon before surgery. SGLT2 inhibitors were routinely continued in the ambulatory setting until admission, after which they were withheld and not prescribed during hospitalization. Consequently, the last dose was typically taken 12 to 24 hours before surgery.

Definition of Ketoacidosis

Postoperative ketoacidosis was defined as the presence of capillary ketones ≥0.6 mmol/L together with metabolic acidosis, established by either arterial pH <7.30 or serum bicarbonate <18 mmol/L[1] in at least 1 postoperative sample. This definition was chosen in line with recent reports in the literature[12] and was prospectively applied to allow detection of early or incipient cases of postoperative ketoacidosis, before progression to more severe metabolic derangements. EuDKA was also considered in those patients with plasma glucose <200 mg/dL.[13]

Preoperative Ketonemia and Fasting Conditions

Preoperative ketonemia measurements were obtained for 116 patients. Five patients (4.3%) showed capillary ketonemia ≥0.6 mmol/L on admission to the cardiothoracic ward the day before surgery and 4 patients (3.4%) on the morning of surgery after overnight fasting. In our institution, elective cardiac surgery patients typically fast for approximately 12 hours between the evening meal before surgery and anesthesia induction. This fasting period was uniform across the cohort and did not induce significant ketosis, as reflected by the low proportion of positive preoperative ketonemia. Although factors such as dehydration or reduced carbohydrate intake may contribute to mild metabolic alterations, their overall impact before surgery appeared negligible in our population.

Postoperative Ketoacidosis Management

All episodes of postoperative ketoacidosis were managed according to a standardized protocol supervised by the on-call intensivist team. Because most cases corresponded to euglycemic ketoacidosis, treatment included continuous infusion of glucose-containing fluids with titration based on blood glucose levels, early initiation of a regular insulin infusion, and frequent monitoring of arterial blood gases and capillary ketonemia until resolution of acidosis. This structured approach, consistently applied by the same ICU team, facilitated early correction and may have contributed to the absence of adverse clinical outcomes.

Statistical Analysis

Quantitative variables were expressed as median and interquartile range (IQR), while qualitative variables were presented as absolute and relative frequencies (n, %). Quantitative variables were analyzed using the Shapiro-Wilk test for normality. Nonparametric tests (Mann-Whitney *U*) were used for group comparisons. Categorical variables were compared using the chi-square or Fisher exact test as appropriate. To address potential confounders, multivariate logistic regression analyses were performed, including clinically relevant variables such as diabetes mellitus, type of surgery, cardiopulmonary bypass (CPB) duration, and chronic kidney disease. The dependent variable was the presence of postoperative ketoacidosis. Results are presented as odds ratios (ORs) with 95% confidence intervals. Analyses were performed using SPSS v26.0 (SPSS, Inc) and Stata 16.0 (StataCorp). A 2-sided p-value <0.05 was considered statistically significant. The advanced statistical analysis, including model estimation and logistic regression, was conducted independently by the clinical research support unit of the hospital's research institute.

Results

Of the 107 patients included in the study, 53% were male, 67% had hypertension, 29% had diabetes, and 59% had dyslipidemia. The median age was 66 years (IQR, 57-73 years), and BMI was 26.4 kg/m^2 (IQR, 24.38-30.16 kg/m^2). The most frequent type of surgery was combined procedures, 37.9% (eg, aortic ± valve, coronary ± valve), followed by coronary artery bypass grafting (17.7%), aortic valve (15.9%), and mitral valve surgery (12.1%). Baseline characteristics such as age, sex, BMI, and prevalence of hypertension and dyslipidemia were comparable between groups (Table 1).

Of the 107 patients included, 26 (24.29%) developed postoperative ketoacidosis. The use of SGLT2 inhibitors was significantly associated with the development of ketoacidosis: 17

Table 1
Baseline Characteristics by the Presence of Ketoacidosis

Variable	No Ketoacidosis (n = 83)	Ketoacidosis (n = 24)	p-value
Age, y	66 [57-73]	65 [58-72]	0.88
Male sex	44 (53)	13 (54)	0.91
BMI, kg/m^2	26.4 [24.4-30.1]	26.6 [24.2-30.3]	0.84
Hypertension	56 (67)	18 (75)	0.45
Diabetes mellitus	16 (19)	14 (58)	0.001
Dyslipidemia	49 (59)	15 (63)	0.73
SGLT2i use	12 (14)	16 (67)	<0.001

Demographic, comorbidity, and preoperative medication data in patients with and without postoperative ketoacidosis. Continuous variables are expressed as median [IQR] and categorical variables as n (%). The p-values correspond to comparisons between groups using the Student *t*, Mann-Whitney, or χ^2 tests as appropriate.
Abbreviations: BMI, body mass index; SGLT2i, sodium-glucose cotransporter 2 inhibitor.

of 31 patients on SGLT2 inhibitors (54.8%) developed ketoacidosis, compared to 9 of 76 patients (11.8%) not on these medications ($p < 0.001$). Diabetes mellitus was also significantly associated with ketoacidosis (OR, 5.13; 95% CI, 1.89-13.91; $p = 0.001$). Mean postoperative glucose concentration at the time of ketoacidosis diagnosis was 174 ± 33 mg/dL compared with 151 ± 32 mg/dL in patients without ketoacidosis ($p = 0.113$), confirming the predominantly euglycemic profile of these episodes.

Analytical differences included lower mean arterial pH in patients with ketoacidosis (7.28 *v* 7.35; $p < 0.001$), lower bicarbonate levels (21.6 *v* 22.8 mmol/L; $p = 0.012$), and a more negative base excess (−4.1 *v* −2.7 mmol/L; $p = 0.03$). Lactate levels were slightly lower in the ketoacidosis group (1.2 *v* 1.8 mmol/L), although this difference was not statistically significant ($p = 0.093$) (Table 2).

No significant differences were observed in surgical variables, such as EUROSCORE II or CPB time (Table 3). The median EUROSCORE II was 2.20 (IQR, 1.30-4.04) in patients with ketoacidosis versus 2.02 (IQR, 1.13-3.10) in those without ($p = 0.24$). Regarding CPB time, more complex or combined procedures, including aortic surgery and pulmonary thromboendarterectomy, were associated with longer CPB times than isolated valve or coronary artery bypass grafting operations, reflecting greater surgical complexity. However, CPB duration was not significantly different between patients with and without postoperative ketoacidosis. The median in those patients with ketoacidosis was 118.50 (IQR, 86.00-164.00) versus 132.00 (IQR, 101.00-194.00) in those without ($p = 0.11$).

Furthermore, clinical outcomes, including ICU length of stay, duration of mechanical ventilation, and in-hospital mortality, did not differ significantly between patients with and without postoperative ketoacidosis. The median ICU stay was 5 days (IQR, 5.0-7.0) in patients with ketoacidosis versus 4.5 days (IQR, 4.0-6.0) in those without ($p = 0.48$). The duration of mechanical ventilation was also comparable (median, 7.50 hours [IQR, 6.0-20.0] *v* 6 hours [IQR, 5.0-21.0]; $p = 0.5$). Regarding postoperative outcomes, the need for surgical reintervention was low, occurring in 7 of 107 patients (6.5%). Specifically, 6 of the 81 patients (7.4%) without ketoacidosis and 1 of the 26 patients (3.8%) with ketoacidosis required reoperation ($p = 0.76$), indicating no significant association between postoperative ketoacidosis and surgical reintervention. Mortality occurred in 1 of 26 patients with ketoacidosis and in 2 of 83 without ($p = 0.29$) (Table 4).

In univariate analysis, SGLT2 inhibitor use was strongly associated with postoperative ketoacidosis (OR, 10.30; 95% CI, 4.08-26.04; $p < 0.001$). After adjustment for diabetes mellitus, the association remained significant (adjusted OR, 6.89; 95% CI, 2.51-18.86; $p < 0.001$).

Adjustment for type of surgery yielded a similar trend (adjusted OR, 15.67; 95% CI, 5.42-45.28; $p < 0.001$), although wide confidence intervals indicated model instability due to small subgroup sizes. Further adjustment for CPB duration and renal function did not materially change the association (adjusted OR ≈ 10). These findings support an independent relationship between perioperative SGLT2 inhibitor use and postoperative ketoacidosis.

Table 2
Postoperative Biochemical Values

Parameter	No Ketoacidosis, Median [IQR]	Ketoacidosis, Median [IQR]	p-value
pH	7.353 [7.34 to 7.36]	7.289 [7.30 to 7.32]	<0.001
Lactate, mmol/L	1.812 [1.61 to 2.03]	1.279 [1.05 to 1.83]	0.093
Base excess, mmol/L	−2.775 [−3.03 to −2.17]	−4.129 [−4.98 to −3.41]	0.030
Bicarbonate, mmol/L	22.844 [22.5 to 23.23]	21.679 [20.9 to 22.25]	0.012

Comparison of arterial blood gas and biochemical parameters between patients with and without postoperative ketoacidosis during the first postoperative hours. The table summarizes acid-base status and metabolic variables, including pH, bicarbonate, base excess, and lactate concentrations, measured according to a predefined sampling protocol.

Table 3
Surgical and Clinical Variables

Variable	No Ketoacidosis (n = 83)	Ketoacidosis (n = 24)	p-value
EUROSCORE II	2.02 [1.13-3.10]	2.20 [1.30-4.04]	0.24
CPB time, min	132.00 [101.00-194.00]	118.50 [86.00-164.00]	0.11
ICU stay, d	4.50 [4.00-6.00]	5.00 [3.00-7.00]	0.48
Mechanical ventilation, h	6.00 [5.00-21.00]	7.50 [6.00-20.00]	0.29
In-hospital mortality	4 (4.94)	0 (0.00)	0.26
Surgical reintervention	6 (7.84)	1 (3.8)	0.76

Comparison of intraoperative characteristics and postoperative outcomes between patients with and without postoperative ketoacidosis. Values are presented as median [IQR] or n (%). Variables include EUROSCORE II, CPB, and postoperative outcomes such as duration of mechanical ventilation, ICU length of stay, surgical reintervention, and in-hospital mortality.
Abbreviations: CPB, cardiopulmonary bypass; ICU, intensive care unit.

Discussion

EuDKA has emerged as an important clinical entity, largely due to the widespread use of SGLT2 inhibitors. These drugs are increasingly used for glycemic control in patients with

Table 4
Association Between Selected Risk Factors and Postoperative Ketoacidosis

Variable	Patients With Ketoacidosis, No. (%)	OR (95% CI)	p-value
SGLT2 inhibitors	16/28 (57.1)	9.04 (3.12-26.13)	<0.001
Diabetes mellitus	14/30 (46.7)	5.13 (1.89-13.91)	0.001

Univariate logistic regression analysis exploring the association of SGLT2 inhibitors and diabetes mellitus with postoperative ketoacidosis.
Abbreviations: OR, odds ratio; SGLT2, sodium-glucose cotransporter-2.

type 2 diabetes by inhibiting renal glucose reabsorption and promoting glucosuria. In addition, they confer cardiovascular benefits, such as weight loss, blood pressure reduction, improved HbA1c levels, increased high-density lipoprotein cholesterol, renal protection (eg, reduced proteinuria), and decreased cardiovascular mortality in patients with heart failure with reduced ejection fraction. This has broadened their indications to include cardiovascular risk reduction in heart failure and chronic kidney disease.

The pathophysiological basis for these findings lies in the SGLT2 inhibitor–induced shift toward ketone body production, further exacerbated by the surgical stress response, fasting, and insulin suppression. The pharmacokinetics of SGLT2 inhibitors reveal a mean elimination half-life of approximately 13 hours for most agents, including empagliflozin and dapagliflozin.[14] While this half-life might suggest a rapid decline in plasma concentration, the pharmacodynamic effects—particularly the inhibition of renal glucose reabsorption and consequent promotion of ketogenesis—may persist beyond the expected duration of measurable drug levels.[15]

In recent years, as the clinical indications for SGLT2 inhibitors have expanded beyond glycemic control to include heart failure and chronic kidney disease, we have increasingly encountered cases of euDKA in our daily clinical practice. Some of these episodes have been of considerable severity, raising concerns about the potential risks associated with these medications in the perioperative setting. These observations led us to question how frequently this complication might occur in our cardiac surgery population and whether it could significantly impact patient outcomes.

In this prospective observational study, we report a 24.29% incidence of postoperative ketoacidosis in patients undergoing elective cardiac surgery, a frequency that is notably higher than expected and suggests that this is an underrecognized complication in this setting. The most significant association was observed with the chronic use of SGLT2 inhibitors, with more than half of treated patients developing ketoacidosis. The definition of ketoacidosis varies slightly across studies, with most guidelines requiring ketosis plus acidosis characterized by either pH <7.30 or bicarbonate <18 mmol/L. Our operational definition prospectively captured ketosis with low pH, in line with previous work, including a recent *Journal of Cardiothoracic and Vascular Anesthesia* report in postcardiac surgery patients treated with SGLT2 inhibitors.[16] We acknowledge that some patients fulfilled our criteria before their bicarbonate had fallen below 18 mmol/L, representing an earlier or less severe stage of ketoacidosis. We believe this approach is clinically relevant, as it allows the identification of metabolic derangements at an incipient phase, when timely recognition and intervention may prevent progression. At the time of data collection, perioperative recommendations regarding SGLT2 inhibitor discontinuation were not yet established, and most patients continued therapy until the day before surgery. This likely contributed to the relatively high incidence of postoperative ketoacidosis observed in our cohort. Current guidelines now recommend withholding these agents for at least 3 days prior to elective procedures, to minimize the risk of euDKA, acknowledging that while the half-life is not excessively long, the metabolic changes induced by the drug warrant a washout period in high-risk situations like surgery. This practice may reduce risk in future cohorts.

Within the SGLT2 inhibitor group, point estimates suggest that the probability of developing postoperative ketoacidosis may be higher in patients taking empagliflozin than in those taking dapagliflozin, although the confidence intervals for both estimates were wide, likely due to the limited number of patients receiving these agents who developed the complication.

Our findings are consistent with previous reports describing postoperative euglycemic ketoacidosis in patients treated with SGLT2 inhibitors. Several retrospective series have reported similar trends, although with considerable methodological variability. For instance, Murugesan et al[17] observed postoperative eDKA in 70.8% of 24 cardiac surgery patients taking SGLT2 inhibitors, while Thiruvenkatarajan et al[18] identified comparable cases in their systematic review. Kim et al[19] and Brekke et al[20] also highlighted a higher prevalence of metabolic acidosis and lower base excess among SGLT2 inhibitor users, suggesting a pathophysiological predisposition in this population.

More recently, Auerbach et al[12] conducted a large retrospective study including 1,654 elective cardiac surgery patients, of whom 53 (3.2%) were receiving SGLT2 inhibitors. Euglycemic ketoacidosis occurred in 8 of these patients (15.1%), and although mortality and overall hospital stay did not differ, those with eDKA experienced longer ICU stays. The broader inclusion criteria and retrospective nature of that study may have led to underdiagnosis of mild or early biochemical cases, as perioperative ketone screening was not systematic. In contrast, our study is the first prospective analysis specifically designed to monitor ketoacidosis after cardiac surgery, with predefined sampling and analytical criteria. This approach likely enabled early detection of incipient ketoacidosis, accounting for the higher incidence observed and the absence of adverse clinical outcomes.

From a preventive perspective, several strategies may help mitigate risk: shortening preoperative fasting, considering carbohydrate loading when appropriate, ensuring adequate perioperative glucose and insulin supplementation, and incorporating routine ketone monitoring in high-risk patients. These measures, alongside appropriate timing of SGLT2 inhibitor discontinuation, may help reduce the incidence and severity of postoperative euglycemic ketoacidosis in cardiac surgery.

In the context of cardiac surgery, patients are particularly vulnerable to the development of complex acid-base disturbances. One of the most feared complications is severe lactic acidosis secondary to low cardiac output syndrome or tissue hypoperfusion, which often occurs in the early postoperative period.[21] Our findings suggest that in patients treated with SGLT2 inhibitors, an additional ketone-driven acidosis may coexist, further aggravating the acid-base imbalance. The coexistence of both lactic and ketoacidosis could not only complicate the metabolic profile but also potentially worsen hemodynamic instability and tissue oxygen delivery.

Analytical differences such as lower arterial pH, reduced bicarbonate, and more negative base excess support the biochemical consistency of these episodes. Interestingly, lactate levels were not always elevated, suggesting that the observed acidosis was not lactic in origin, which reinforces the diagnosis of ketoacidosis. These data reinforce that the metabolic disturbances observed were consistent with euglycemic ketoacidosis, a phenomenon characteristic of SGLT2 inhibitor therapy. The lack of a significant increase in glucose and the slightly lower lactate levels among patients with ketoacidosis suggest that these episodes occurred in the context of preserved tissue perfusion and effective hemodynamic management. This supports the notion that early detection and treatment prevented the development of severe or clinically relevant metabolic decompensation.

From a clinical outcome perspective, our data did not show significant differences in ICU stay, ventilation time, need for surgical reintervention, or mortality between patients with and without ketoacidosis. The design of the present study poses inherent limitations in assessing the true clinical impact of ketoacidosis in this population. This study was conducted in a tertiary cardiac ICU staffed by intensivists trained in the comprehensive management of critically ill patients, including endocrine and metabolic disorders. Early detection of metabolic alterations and prompt initiation of insulin infusion, glucose control, and correction of acid-base and electrolyte imbalances are integral parts of routine postoperative care in our unit. These practices likely explain why, despite a relatively high incidence, most episodes were mild, transient, and rapidly reversible, without a significant impact on postoperative outcomes such as ICU stay, mechanical ventilation, or mortality. The observed biochemical derangements—lower pH, bicarbonate, and base excess—were therefore clinically contained through early intervention. We believe that such proactive management, supported by standardized monitoring protocols, is key to preventing progression to severe or clinically overt ketoacidosis. Without these measures, ketoacidosis would likely be recognized later and in a more advanced stage, potentially leading to delayed recovery, increased need for organ support, and worse outcomes.

While most episodes of postoperative ketoacidosis identified in this cohort were mild and promptly corrected, subclinical or transient metabolic disturbances may still exert downstream effects on recovery, such as transient fluid overload, impaired wound healing, or delayed extubation. These subtle consequences might remain clinically inapparent when the disorder is detected early and treated effectively, as in our setting.

Our study has several limitations. The sample size is relatively modest and precluded multivariate analysis. The structured sampling protocol employed in this study enabled systematic and early detection of postoperative ketoacidosis through standardized measurement of capillary ketones and arterial blood gases at predefined time points. This approach allowed the identification of early or mild metabolic derangements that may precede clinically overt ketoacidosis. However, because sampling was intermittent, transient or short-lived fluctuations in ketone levels or acid-base status may have gone undetected. Continuous or more frequent measurements could potentially identify additional brief episodes, although their clinical relevance remains uncertain. Overall, our standardized and reproducible sampling strategy proved effective in capturing the cases of greatest clinical significance during the immediate postoperative period. Another limitation of our study is the lack of systematically collected intraoperative glucose data, including glucose trajectories and insulin or dextrose administration during surgery. This omission precludes a more detailed exploration of the metabolic drivers of postoperative ketoacidosis during the intraoperative phase. Nevertheless, the strong association we observed between SGLT2 inhibitor therapy and postoperative ketoacidosis, despite standardized intraoperative glucose management, supports the robustness of our findings. Future studies should incorporate detailed intraoperative glucose monitoring to better elucidate the interplay between perioperative glucose control, SGLT2 inhibitor therapy, and the risk of ketoacidosis.

Multivariate logistic regression confirmed that SGLT2 inhibitor use was independently associated with postoperative ketoacidosis, even after adjustment for diabetes, renal function, type of surgery, and CPB duration. However, wide confidence intervals and some model instability reflect the limited sample size, underscoring the need for larger multicenter studies to refine these estimates and explore potential effect modifiers. Finally, the single-center design may limit generalizability.

Conclusions

Postoperative ketoacidosis in cardiac surgery patients is more frequent than previously recognized and appears strongly associated with the use of SGLT2 inhibitors. Given the clinical implications of this finding, perioperative management strategies should be reassessed, and blood ketone monitoring protocols must be considered in this population.

Further multicenter studies are needed to validate these findings and guide evidence-based recommendations on SGLT2 inhibitor discontinuation in cardiac surgical patients.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

José Ginestal-Calvo: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **J.L. Flordelís-Lasierra:** Validation, Supervision, Project administration, Methodology, Investigation, Conceptualization. **J.L. Pérez-Vela:** Validation, Supervision, Project administration, Methodology, Investigation, Conceptualization. **Laura Campos-Bermejo:** Project administration, Investigation, Data curation. **Cristian Coso-Garrido:** Resources, Project administration, Data curation. **Luis Blanco-Carrasco:** Investigation,

Data curation. **Carmen Galiano-Carrilero:** Methodology, Data curation. **Helena Domínguez-Aguado:** Supervision, Methodology, Investigation. **Emilio Renes-Carreño:** Supervision, Software, Methodology.

References

1 Nasa P, Chaudhary S, Shrivastava PK, Singh A. Euglycemic diabetic ketoacidosis: a missed diagnosis. World J Diabetes 2021;12(5):514–23.
2 Long B, Lentz S, Koyfman A, Gottlieb M. Euglycemic diabetic ketoacidosis: etiologies, evaluation, and management. Am J Emerg Med 2021;44:157–60.
3 Koceva A, Kravos Tramšek NA. From sweet to sour: SGLT-2-inhibitor-induced euglycemic diabetic ketoacidosis. J Pers Med 2024;14(7):665.
4 Goldenberg RM, Berard LD, Cheng AYY, et al. SGLT2 inhibitor–associated diabetic ketoacidosis: clinical review and recommendations for prevention and diagnosis. Clin Ther 2016;38:2654–64;e1.
5 Watson K. Surgical risk in patients with metabolic syndrome: Focus on lipids and hypertension. Curr Cardiol Rep 2006;8(6):433–8.
6 Jakob SM, Ensinger H, Takala J. Metabolic changes after cardiac surgery. Curr Opin Clin Nutr Metab Care 2001;4(2):149–55.
7 Lamperti M, Romero CS, Guarracino F, et al. Preoperative assessment of adults undergoing elective noncardiac surgery: updated guidelines from the European Society of Anaesthesiology and Intensive Care. Eur J Anaesthesiol 2025;42:1–35.
8 FDA. Drug safety communication: FDA warns that SGLT2 inhibitors for diabetes may result in a serious condition of too much acid in the blood. May 15, 2015. Revised March 15, 2022. Available at: https://www.fda.gov/drugs/drug-safety-and-availability/fda-drug-safety-communication-fda-warns-sglt2-inhibitors-diabetes-may-result-serious-condition-too. Accessed May 3, 2025.
9 Galindo RJ, Fayfman M, Umpierrez GE. Perioperative management of hyperglycemia and diabetes in cardiac surgery patients. Endocrinol Metab Clin North Am 2018;47:203–22.
10 Ginestal J, Flordelís JL, Pérez-Vela JL. Sodium-glucose cotransporter-2 inhibitors and the significant risk of euglycaemic keto-acidosis in cardiac surgery. Eur J Anaesthesiol 2025;42(6):571–2.
11 Vandenbroucke JP, von Elm E, Altman DG, et al. Strengthening the Reporting of Observational Studies in Epidemiology (STROBE): explanation and elaboration. PLoS Med 2007;4:e297.
12 Auerbach JS, Gershengorn HB, Aljure OD, et al. Postcardiac surgery euglycemic diabetic ketoacidosis in patients on sodium-glucose cotransporter 2 inhibitors. J Cardiothorac Vasc Anesth 2023;37(6):956–63.
13 Umpierrez GE, Davis GM, ElSayed NA, et al. Hyperglycemic crises in adults with diabetes: a consensus report. Diabetes Care 2024;47(8):1257–75.
14 Weil RM. Getting to know ketones. Diabetes Self Manag 2003;20(6):100–1;103-4.
15 Garcia-Ropero A, Badimon JJ, Santos-Gallego CG. The pharmacokinetics and pharmacodynamics of SGLT2 inhibitors for type 2 diabetes mellitus: the latest developments. Exp Opin Drug Metab Toxicol 2018;14(12):1287–302.
16 Pharmacokinetics Scheen AJ. Pharmacodynamics and clinical use of SGLT2 inhibitors in patients with type 2 diabetes mellitus and chronic kidney disease. Clin Pharmacokinet 2015;54(7):691–708.
17 Murugesan KB, Balakrishnan S, Arul A, et al. A retrospective analysis of the incidence, outcome and factors associated with the occurrence of euglycemic ketoacidosis in diabetic patients on sodium glucose co-transporter–2 inhibitors undergoing cardiac surgery. Ann Card Anaesth 2022;25:460–5.
18 Thiruvenkatarajan V, Meyer EJ, Nanjappa N, et al. Perioperative diabetic ketoacidosis associated with sodium-glucose co-transporter-2 inhibitors: a systematic review. Br J Anaesth 2019;123:27–36.
19 Kim HA, Kim JY, Kim YH, Lee YT, Park PW. Missed postoperative metabolic acidosis associated with sodium-glucose transporter 2 inhibitors in cardiac surgery patients: a retrospective analysis. Sci Rep 2024;14(1):8087.
20 Brekke HK, Holmaas G, Astor MC, et al. Metabolic acidosis in patients with diabetes 2 undergoing cardiac surgery: the impact of SGLT2 inhibitor use: a retrospective cohort study. Eur J Anaesthesiol 2025;42(2):152–61.
21 Pérez Vela JL, Martín Benítez JC, Carrasco González M, et al. Clinical practice guide for the management of low cardiac output syndrome in the postoperative period of heart surgery. Med Intens 2012;36(4):e1–44.